

21. Let $y = y(x)$ be the solution curve of the differential equation, $(y^2 - x) \frac{dy}{dx} = 1$, satisfying $y(0) = 1$. This curve intersects the x -axis at a point whose abscissa is :

माना अवकल समीकरण $(y^2 - x) \frac{dy}{dx} = 1$ का हल वक्र $y = y(x)$, $y(0) = 1$ को सन्तुष्ट करता है। यह वक्र x -अक्ष को जिस बिन्दु पर काटता है उसका भुज है :

- (A) $2 - e$ (B) 2 (C) $2 + e$ (D) $-e$

Ans. A

Sol. $(y^2 - x) \frac{dy}{dx} = 1$

$$\frac{dx}{dy} + x = y^2$$

Linear differential equation

I.F. = e^y

$$\int d(xe^y) = \int y^2 e^y dy$$

$$xe^y = y^2 e^y - 2ye^y + 2e^y + c$$

Put $x = 0$, $y = 1$ then $c = -e$

$$e^y x = e^y (y^2 - 2y + 2) - e$$

$$x = (y - 1)^2 + 1 - e^{1-y}$$

Put $y = 0 \Rightarrow x = 2 - e$

22. Let T and C respectively be the transverse and conjugate axes of the hyperbola $16x^2 - y^2 + 64x + 4y + 44 = 0$. Then the area of the region above the parabola $x^2 = y + 4$, below the transverse axis T and on the right of the conjugate axis C is :

माना अतिपरवलय $16x^2 - y^2 + 64x + 4y + 44 = 0$ के अनुप्रस्थ तथा संयुग्मी अक्ष क्रमशः T तथा C हैं। तो परवलय $x^2 = y + 4$ के ऊपर, अनुप्रस्थ अक्ष T के नीचे तथा संयुग्मी अक्ष की दांयी तरफ के क्षेत्र का क्षेत्रफल है :

- (A) $4\sqrt{6} + \frac{44}{3}$ (B) $4\sqrt{6} + \frac{28}{3}$ (C) $4\sqrt{6} - \frac{44}{3}$ (D) $4\sqrt{6} - \frac{28}{3}$

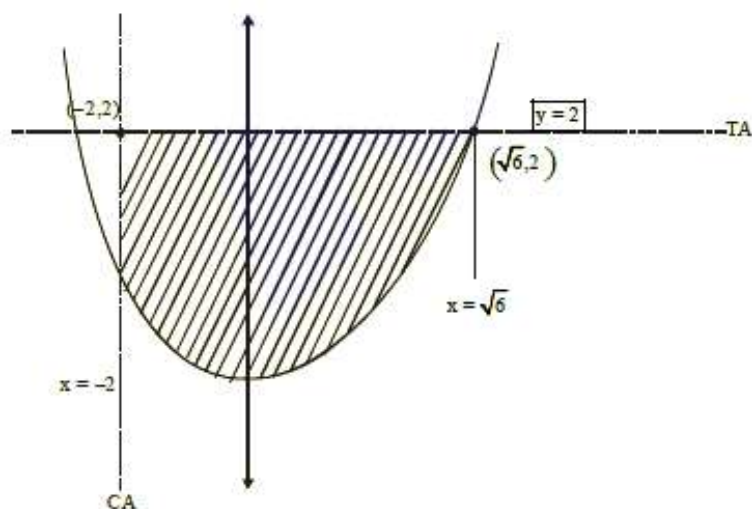
Ans. B

Sol. $16(x^2 + 4x) - (y^2 - 4y) + 44 = 0$

$$16(x + 2)^2 - 64 - (y - 2)^2 + 4 + 44 = 0$$

$$16(x + 2)^2 - (y - 2)^2 = 16$$

$$\frac{(x+2)^2}{1} - \frac{(y-2)^2}{16} = 1$$



$$A = \int_{-2}^{\sqrt{6}} (2 - (x^2 - 4)) dx$$

$$A = \int_{-2}^{\sqrt{6}} (6 - x^2) dx = \left(6x - \frac{x^3}{3} \right)_{-2}^{\sqrt{6}}$$

$$A = \left(6\sqrt{6} - \frac{6\sqrt{6}}{3} \right) - \left(-12 + \frac{8}{3} \right)$$

$$A = \frac{12\sqrt{6}}{3} + \frac{28}{3}$$

$$A = 4\sqrt{6} + \frac{28}{3}$$

23. Consider the differential equation, $y^2 dx + \left(x - \frac{1}{y} \right) dy = 0$. If value of y is 1 when $x = 1$, then the value of x for with $y = 2$, is:

अवकल समीकरण $y^2 dx + \left(x - \frac{1}{y} \right) dy = 0$ पर विचार कीजिए। यदि $x = 1$ पर y का मान 1 है, तो x का मान, जिसके लिए

$y = 2$ है, है:

(A) $\frac{5}{2} + \frac{1}{\sqrt{e}}$

(B) $\frac{3}{2} - \sqrt{e}$

(C) $\frac{3}{2} - \frac{1}{\sqrt{e}}$

(D) $\frac{1}{2} + \frac{1}{\sqrt{e}}$

Ans. C

sol. $y^2 dx + \left(x - \frac{1}{y}\right) dy = 0$

$$\frac{dx}{dy} + \left(\frac{1}{y^2}\right)x = \frac{1}{y^3}$$

$$\text{I.F.} = e^{\int \frac{1}{y^2} dy} = e^{-\frac{1}{y}}$$

its solution is

$$x \cdot e^{-\frac{1}{y}} = \int e^{-\frac{1}{y}} \frac{1}{y^3} dy + c$$

$$\text{put } -\frac{1}{y} = t \Rightarrow \frac{1}{y^2} dy = dt$$

$$\Rightarrow x \cdot e^{-\frac{1}{y}} = -\int t e^t dt + c = -t e^t + e^t + c$$

$$x \cdot e^{-\frac{1}{y}} = e^{-\frac{1}{y}} \left(\frac{1}{y} + 1\right) + c \text{ passes through } (1, 1)$$

$$\Rightarrow 1 = 2 + ce \Rightarrow c = -\frac{1}{e}$$

$$\Rightarrow x \left(1 + \frac{1}{y}\right) - \frac{1}{e} e^{\frac{1}{y}} \text{ passes through } (k, 2)$$

$$\Rightarrow k = \frac{3}{2} - \frac{1}{\sqrt{e}}$$

24. Let slope of the tangent line to a curve at any point $P(x, y)$ be given by $\frac{xy^2 + y}{x}$. If the curve intersects the line $x + 2y = 4$ at $x = -2$, then the value of y , for which the point $(3, y)$ lies on the curve, is :

माना एक वक्र के किसी बिंदु $P(x, y)$ पर स्पर्श रेखा की प्रवणता $\frac{xy^2 + y}{x}$ द्वारा दी गई है। यदि यह वक्र, रेखा

$x + 2y = 4$ को $x = -2$ पर काटता है, तो y का वह मान, जिसके लिए बिंदु $(3, y)$ वक्र पर है, है :

- (A) $-\frac{18}{19}$ (B) $\frac{18}{35}$ (C) $-\frac{18}{11}$ (D) $-\frac{4}{3}$

Ans. A

Sol. $\frac{dy}{dx} = \frac{xy^2 + y}{x}$

$$\Rightarrow \frac{dy}{dx} = y^2 + \frac{1}{x}y$$

$$\Rightarrow \frac{1}{y^2} \frac{dy}{dx} - \frac{1}{y} \cdot \frac{1}{x} = 1$$

put $-\frac{1}{y} = z$

$$\frac{1}{y^2} \frac{dy}{dx} = \frac{dz}{dx}$$

$$\Rightarrow \frac{dz}{dx} + z \cdot \frac{1}{x} = 1$$

$$\text{I.F} = e^{\int \frac{1}{x} dx} = x$$

$$\Rightarrow z \cdot x = \int x \cdot 1 dx + c$$

$$\Rightarrow z \cdot x = \frac{x^2}{2} + c$$

$$\Rightarrow \therefore z = \frac{x}{2} + \frac{c}{x}$$

$$\therefore -\frac{1}{y} = \frac{x}{2} + \frac{c}{x} = \frac{x^2 + 2c}{2x}$$

$$\therefore y = \frac{-2x}{x^2 + 2c} \quad \dots\dots\dots(i)$$

$\therefore$ Curve intersect the line $x + 2y = 4$ at $x = -2$

i.e. $-2 + 2y = 4 \quad \therefore y = 3$

$\therefore (-2, 3)$ lies on the curve

$$\therefore 3 = \frac{-2 \times (-2)}{4 + 2c}$$

$$\Rightarrow 12 + 6c = 4$$

$$6c = -8 \quad \therefore c = -\frac{4}{3}$$

The equation of the required curve is

$$y = \frac{-2x}{x^2 - \frac{8}{3}}$$

$$\therefore y = \frac{6x}{8 - 3x^2}$$

when $x = 3$

$$y = \frac{18}{8 - 27} \quad \therefore y = -\frac{18}{19}$$

25. Consider the integral $I = \int_0^{10} \frac{[x]e^{[x]}}{e^{x-1}} dx$ where $[x]$ denotes the greatest integer less than or equal to x . Then the value of I is equal to:

समाकलन $I = \int_0^{10} \frac{[x]e^{[x]}}{e^{x-1}} dx$ का विचार कीजिए, जहाँ $[x]$ महत्तम पूर्णांक $\leq x$ है। तो I का मान बराबर है :

- (A) $45(e - 1)$ (B) $45(e + 1)$ (C) $9(e + 1)$ (D) $9(e - 1)$

Ans. A

Sol. $x \frac{dv}{dx} = \frac{-1 - v^2}{2}$

$$I = \left[e \int_0^1 \frac{0}{e^{\{x\}}} dx + \int_1^2 \frac{1}{e^{\{x\}}} dx + \int_2^3 \frac{2}{e^{\{x\}}} dx + \dots + \int_9^{10} \frac{9}{e^{\{x\}}} dx \right]$$

$$I = e(0 + 1 + 2 + \dots + 9) \int_0^1 \frac{dx}{e^x}$$

$$I = 45e \int_0^1 e^{-x} dx$$

$$I = 45e \left[-e^{-x} \right]_0^1$$

$$I = 45e \left[-e^{-1} + 1 \right]$$

$$I = 45(e - 1)$$

26. Let a function $f: [0, 5] \rightarrow \mathbf{R}$ be continuous, $f(1) = 3$ and F be defined as :

$$F(x) = \int_1^x t^2 g(t) dt, \text{ where } g(t) = \int_1^t f(u) du.$$

Then for the function F , the point $x = 1$ is :

- (A) not a critical point (B) a point of local maxima
(C) a point of inflection (D) a point of local minima

माना एक फलन $f: [0, 5] \rightarrow \mathbf{R}$ संतत है, $f(1) = 3$ है तथा $F, F(x) = \int_1^x t^2 g(t) dt$ द्वारा परिभाषित है, जहाँ

$g(t) = \int_1^t f(u) du$ है तो फलन F के लिए, बिन्दु $x = 1$ एक :

- (A) क्रांतिक बिन्दु नहीं है। (B) स्थानीय उच्चिष्ठ बिन्दु है।
(C) नति परिवर्तन बिन्दु है। (D) स्थानीय निम्ननिष्ठ बिन्दु है।

Ans. D

Sol. $f: [0, 5] \rightarrow \mathbf{R} \quad f(1) = 3$

$$F(x) = \int_1^x t^2 g(t) dt$$

$$F'(x) = x^2 g(x)$$

$$F'(1) = g(1)$$

$$g(t) = \int_1^t f(u) du \quad \{g(1) = 0\}$$

$$F'(1) = g(1) = 0$$

$$F''(x) = x^2 g'(x) + 2xg(x)$$

$$g'(x) = f(x)$$

$$F''(x) = x^2 f(x) + 2xg(x)$$

$$F''(1) = f(1) + 2g(1) = 3$$

From (1) and (2), $F(x)$ has local minimum at $x = 1$

27. Let $y = y(x)$ be a solution curve of the differential equation $(1 - x^2 y^2) dx = y dx + x dy$.

If the line $x = 1$ intersects the curve $y = y(x)$ at $y = 2$ and the line $x = 2$ intersects the curve $y = y(x)$ at

$y = \alpha$, then a value of α is :

माना अवकल समीकरण $(1 - x^2y^2)dx = ydx + xdy$ का हल वक्र $y = y(x)$ है। यदि रेखा $x = 1$, वक्र $y = y(x)$ को $y = 2$ पर काटती है तथा रेखा $x = 2$, वक्र $y = y(x)$ को $y = \alpha$ पर काटती है, तो α का एक मान है

- (A) $\frac{3e^2}{2(3e^2+1)}$ (B) $\frac{3e^2}{2(3e^2-1)}$ (C) $\frac{1+3e^2}{2(3e^2-1)}$ (D) $\frac{1-3e^2}{2(3e^2+1)}$

Ans. C

Sol. $dx = \frac{d(xy)}{1-(xy)^2} \Rightarrow 2dx = \frac{d(xy)}{1-xy} + \frac{d(xy)}{1+xy}$

$$2x + c = \ln \left| \frac{1+xy}{1-xy} \right| \Rightarrow \left| \frac{xy+1}{xy-1} \right| = e^c \cdot e^{2x}$$

$$\because y(1) = 2$$

$$\Rightarrow 3 = e^c \cdot e^2$$

$$\Rightarrow e^c = \frac{3}{e^2} e^{2x}$$

So at $x = 2$

$$\Rightarrow \left| \frac{2\alpha+1}{2\alpha-1} \right| = 3e^2$$

Case : 1 for +ve sign

$$\Rightarrow \frac{2\alpha+1}{2\alpha-1} = \frac{3e^2}{1}$$

$$\Rightarrow \frac{4\alpha}{2} = \frac{3e^2+1}{3e^2-1}$$

$$\Rightarrow \alpha = \frac{1}{2} \left(\frac{3e^2+1}{3e^2-1} \right)$$

Case: 2 for (-ve)sign not required

28. If $y = y(x)$ is the solution of the differential equation, $e^y \left(\frac{dy}{dx} - 1 \right) = e^x$ such that $y(0) = 0$, then $y(1)$ is equal to:

यदि अवकलन समीकरण, $e^y \left(\frac{dy}{dx} - 1 \right) = e^x$, जबकि $y(0) = 0$, का हल $y = (x)$ है, $y(1)$ बराबर है:

- (A) $\log_e 2$ (B) $2 + \log_e 2$ (C) $1 + \log_e 2$ (D) $2e$

Ans. C

Sol. $c^y = t \quad e^y \frac{dy}{dx} = \frac{dt}{dx}$

So $\frac{dt}{dx} - t = e^x$

Integrating factor = $e^{-\int 1 dx} = e^{-x}$

So $\int \frac{d}{dx}(e^{-x} \cdot t) = \int 1 dx$

$\Rightarrow t \cdot e^{-x} = x + c$

$\Rightarrow t = x e^x + c e^x$

$\Rightarrow e^y = x e^x + c e^x$

Since $y(0) = 0$

$\Rightarrow c = 1$

So $e^y = x e^x + e^x$

$y(1) = \log_e(e^x(x+1)) \big|_{x=1} = 1 + \log_e 2$

29. If the solution curve $f(x, y) = 0$ of the differential equation $(1 + \log_e x) \frac{dx}{dy} - x \log_e x = e^y, x > 0$, passes through the points $(1, 0)$ and $(\alpha, 2)$ then α^α is equal to :

यदि अवकल समीकरण $(1 + \log_e x) \frac{dx}{dy} - x \log_e x = e^y, x > 0$ का हल वक्र $f(x, y) = 0$ बिंदुओं $(1, 0)$ तथा $(\alpha, 2)$ से होकर

जाता है, तो α^α बराबर है :

(A) e^{2e^2}

(B) e^{e^2}

(C) $e^{2e^{\sqrt{2}}}$

(D) $e^{\sqrt{2}e^2}$

Ans. A

Sol. $(1 + \ln x) \frac{dx}{dy} - x \ln x = e^y$

Let $x \ln x = t$

$(1 + \ln x) \frac{dx}{dy} = \frac{dt}{dy}$

$\frac{dt}{dy} - t = e^y$

If $= e^{\int -dy} = e^{-y}$

$t \cdot e^{-y} = \int e^y e^{-y} dy + c$

$$te^{-y} = y + c$$

$$x \ln xe^{-y} = y + c$$

$$x \ln x = ye^y + ce^y$$

$$(1, 0)0 = C$$

$$\Rightarrow x \ln x = ye^y$$

$$\Rightarrow \alpha \ln \alpha = 2e^2$$

$$\alpha^\alpha = e^{2e^2}$$

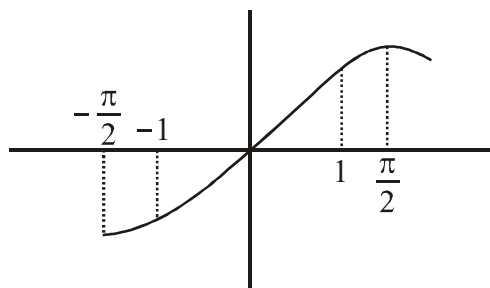
30. The value of $\int_{-\pi/2}^{\pi/2} \frac{dx}{[x] + [\sin x] + 4}$, where $[t]$ denotes the greatest integer less than or equal to t , is :

$\int_{-\pi/2}^{\pi/2} \frac{dx}{[x] + [\sin x] + 4}$, का मान जहाँ $[t]$ महत्तम पूर्णांक है जो t से कम या उसके बराबर है

- (A) $\frac{1}{12}(7\pi - 5)$ (B) $\frac{3}{10}(4\pi - 3)$ (C) $\frac{3}{20}(4\pi - 3)$ (D) $\frac{1}{12}(7\pi + 5)$

Ans. C

S. $\int_{-\pi/2}^{\pi/2} \frac{dx}{[x] + [\sin x] + 4}$



$$\int_{-\pi/2}^{-1} \frac{dx}{-2-1+4} + \int_{-1}^0 \frac{dx}{-1-1+4} + \int_0^1 \frac{dx}{0+0+4} + \int_1^{\pi/2} \frac{dx}{1+0+4}$$

$$= \int_{-\pi/2}^{-1} \frac{dx}{1} + \int_{-1}^0 \frac{dx}{2} + \int_0^1 \frac{dx}{4} + \int_1^{\pi/2} \frac{dx}{5}$$

$$= \left[-1 + \frac{\pi}{2} \right] + \frac{1}{2}[0+1] + \frac{1}{4}[1-0] + \frac{1}{5} \left[\frac{\pi}{2} - 1 \right]$$

$$= \frac{3}{20}(4\pi - 3)$$

31. Let $I = \int_a^b (x^4 - 2x^2) dx$. If I is minimum then the ordered pair (a, b) is :

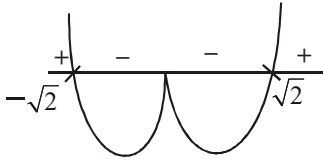
माना $I = \int_a^b (x^4 - 2x^2) dx$ है। यदि I न्यूनतम है, तो क्रमित युग्म (a, b) है :

- (A) $(\sqrt{2}, -\sqrt{2})$ (B) $(0, \sqrt{2})$ (C) $(-\sqrt{2}, \sqrt{2})$ (D) $(-\sqrt{2}, 0)$

Ans. C

Sol. $I = \int_a^b (x^4 - 2x^2) dx$

$$f(x) = x^2(x^2 - 2)$$



for I min $(a, b) = (-\sqrt{2}, \sqrt{2})$

$$I = \int_{-\sqrt{2}}^{\sqrt{2}} (x^4 - 2x^2) dx$$

32. If the integral $\int_0^{10} \frac{[\sin 2\pi x]}{e^{x-[x]}} dx = \alpha e^{-1} + \beta e^{-\frac{1}{2}} + \gamma$, where α, β, γ are integers and $[x]$ denotes the greatest integer less than or equal to x , then the value of $\alpha + \beta + \gamma$ is equal :

यदि समाकलन $\int_0^{10} \frac{[\sin 2\pi x]}{e^{x-[x]}} dx = \alpha e^{-1} + \beta e^{-\frac{1}{2}} + \gamma$ हैं, जहाँ α, β, γ पूर्णांक हैं तथा $[x]$ महत्तम पूर्णांक $\leq x$ हैं, तो

$\alpha + \beta + \gamma$ का मान बराबर है :

- (A) 0 (B) 20 (C) 25 (D) 10

Ans. A

Sol. $I = \int_0^{10} \frac{[\sin 2\pi x]}{e^{x-[x]}} dx$

$$I = \int_0^{10 \times 1} [\sin 2\pi x] e^{-\{x\}} dx$$

$$I = 10 \int_0^1 [\sin 2\pi x] e^{-x} dx$$

$$I = 10 \int_0^1 e^{-x} [\sin 2\pi x] dx$$

$$I = 10 \int_0^{\frac{1}{2}} e^{-x} \times 0 dx + 10 \int_{\frac{1}{2}}^1 e^{-x} (-1) dx$$

$$I = -10 \left(\frac{e^{-x}}{-1} \right) \Big|_{\frac{1}{2}}^1$$

$$I = 10 \left(e^{-1} - e^{-\frac{1}{2}} \right) = \alpha e^{-1} + \beta e^{-\frac{1}{2}} + \gamma$$

$$\alpha = 10$$

$$\beta = -10$$

$$\gamma = 0$$

$$\alpha + \beta + \gamma = 0$$

33. Let $\alpha > 0$. If $\int_0^\alpha \frac{x}{\sqrt{x+\alpha}-\sqrt{x}} dx = \frac{16+20\sqrt{2}}{15}$, then α is equal to :

माना $\alpha > 0$ है। यदि $\int_0^\alpha \frac{x}{\sqrt{x+\alpha}-\sqrt{x}} dx = \frac{16+20\sqrt{2}}{15}$ है, तो α बराबर है :

(A) $\sqrt{2}$

(B) 4

(C) $2\sqrt{2}$

(D) 2

Ans. D

Sol. After rationalising

$$\int_0^\alpha \frac{x}{\alpha} (\sqrt{x+\alpha} + \sqrt{x})$$

$$\int_0^\alpha \frac{1}{\alpha} [(x+\alpha)^{3/2} - \alpha(x+\alpha)^{1/2} + x^{3/2}]$$

$$\frac{1}{\alpha} \left[\frac{2}{5} (x+\alpha)^{5/2} - \alpha \frac{2}{3} (x+\alpha)^{3/2} + \frac{2}{5} x^{5/2} \right] \Big|_0^\alpha$$

$$= \frac{1}{\alpha} \left(\frac{2}{5} (2\alpha)^{5/2} - \frac{2\alpha}{3} (2\alpha)^{3/2} + \frac{2}{5} \alpha^{5/2} - \frac{2}{5} \alpha^{5/2} + \frac{2}{3} \alpha^{5/2} \right)$$

$$\begin{aligned}
 &= \frac{1}{\alpha} \left(\frac{2^{7/2} \alpha^{5/2}}{5} - \frac{2^{5/2} \alpha^{5/2}}{3} + \frac{2}{3} \alpha^{5/2} \right) \\
 &= \alpha^{3/2} \left(\frac{2^{7/2}}{5} - \frac{2^{5/2}}{3} + \frac{2}{3} \right) \\
 &= \frac{\alpha^{3/2}}{15} (24\sqrt{2} - 20\sqrt{2} + 10) = \frac{\alpha^{3/2}}{15} (4\sqrt{2} + 10)
 \end{aligned}$$

$$\text{Now, } \frac{\alpha^{3/2}}{15} (4\sqrt{2} + 10) = \frac{16 + 20\sqrt{2}}{15}$$

$$\Rightarrow \alpha = 2$$

34. The integral $\int \frac{(2x-1) \cos \sqrt{(2x-1)^2 + 5}}{\sqrt{4x^2 - 4x + 6}} dx$ is equal to :

(where c is a constant of integration)

समाकलन $\int \frac{(2x-1) \cos \sqrt{(2x-1)^2 + 5}}{\sqrt{4x^2 - 4x + 6}} dx$ बराबर है :

(जहाँ c एक समाकलन अचर)

(A) $\frac{1}{2} \sin \sqrt{(2x-1)^2 + 5} + c$

(B) $\frac{1}{2} \cos \sqrt{(2x+1)^2 + 5} + c$

(C) $\frac{1}{2} \cos \sqrt{(2x-1)^2 + 5} + c$

(D) $\frac{1}{2} \sin \sqrt{(2x-1)^2 + 5} + c$

Ans. A

Sol. $(2x-1) = t$; $2dx = dt$

$$I = \int \frac{t \cos \sqrt{t^2 + 5}}{t^2 + 5} \frac{dt}{2}$$

$$t^2 + 5 = u^2$$

$$\Rightarrow 2t dt = 2u du$$

$$\Rightarrow I = \int \frac{\cos u \cdot u du}{2u}$$

$$= \frac{1}{2} \sin u + c = \frac{1}{2} \sin \sqrt{t^2 + 5} + C$$

$$= \frac{1}{2} \sin \sqrt{(2x-1)^2 + 5} + C$$

35. The integral $\int \frac{e^{3 \log_e 2x} + 5e^{2 \log_e 2x}}{e^{4 \log_e x} + 5e^{3 \log_e x} - 7e^{2 \log_e x}} dx$, $x > 0$ is equal to : (where c is a constant of integration)

समाकलन $\int \frac{e^{3 \log_e 2x} + 5e^{2 \log_e 2x}}{e^{4 \log_e x} + 5e^{3 \log_e x} - 7e^{2 \log_e x}} dx$, $x > 0$, बराबर है : (जहाँ c एक समाकलन अचर है)

(A) $\log_e \sqrt{x^2 + 5x - 7} + c$

(B) $\log_e |x^2 + 5x - 7| + c$

(C) $\frac{1}{4} \log_e |x^2 + 5x - 7| + c$

(D) $4 \log_e |x^2 + 5x - 7| + c$

Ans. D

Sol.
$$\int \frac{8x^3 + 20x^2}{x^4 + 5x^3 - 7x^2} dx$$

$$= 4 \int \frac{2x + 5}{x^2 + 5x - 7} dx$$

$$= 4 \ln |x^2 + 5x - 7| + C$$

36. If $\int \frac{dx}{x^3(1+x^6)^{2/3}} = xf(x)(1+x^6)^{1/3} + C$ where C is a constant of integration, then the function $f(x)$ is equal to:

यदि $\int \frac{dx}{x^3(1+x^6)^{2/3}} = xf(x)(1+x^6)^{1/3} + C$ जहाँ C एक समाकलन अचर है, तो फलन $f(x)$ बराबर है :

(A) $-\frac{1}{2x^3}$

(B) $\frac{3}{x^2}$

(C) $-\frac{1}{6x^3}$

(D) $-\frac{1}{2x^2}$

Ans. A

sol.
$$I = \int \frac{dx}{x^3(1+x^6)^{2/3}} = \int \frac{dx}{x^7(1+x^{-6})^{2/3}}$$

Put $1 + x^{-6} = t^3$

$\Rightarrow -6x^{-7}dx = 3t^2 dt$

$\Rightarrow \frac{dx}{x^7} = -\frac{1}{2} t^2 dt$

$$\begin{aligned}
 \Rightarrow I &= \int -\frac{1}{2} \frac{t^2 dt}{t^2} \\
 &= -\frac{1}{2} t + C \\
 &= -\frac{1}{2} (1+x^{-6})^{\frac{1}{3}} + C \\
 &= -\frac{1}{2} \frac{(1+x^{-6})^{\frac{1}{3}}}{x^2} + C \\
 &= -\frac{1}{2x^3} x(1+x^{-6})^{\frac{1}{3}} + C \\
 \Rightarrow f(x) &= -\frac{1}{2x^3}
 \end{aligned}$$

37. If a curve passes through the origin and the slope of the tangent to it at any point (x, y) is $\frac{x^2 - 4x + y + 8}{x - 2}$, then this curve also passes through the point :

यदि एक वक्र मूलबिंदू से होकर जाता है तथा इसके किसी बिंदु (x, y) पर स्पर्श रेखा की प्रवणता $\frac{x^2 - 4x + y + 8}{x - 2}$ है, तो यह वक्र निम्न में से किस बिंदु से भी होकर जाता है ?

- (A) (4, 5) (B) (5, 5) (C) (5, 4) (D) (4, 4)

Ans. B

Sol. $\frac{dy}{dx} = \frac{x^2 - 4x + y + 8}{x - 2}$

$$\frac{dy}{dx} = \frac{(x-2)^2 + y + 4}{(x-2)}$$

Let, $x - 2 = X$

$$y + 4 = Y \quad \left[\frac{dY}{dX} = \frac{dy}{dx} \right]$$

$$\frac{dY}{dX} = \frac{X^2 + Y}{X}$$

$$\frac{dY}{dX} = Y \left(\frac{-1}{X} \right) = -\frac{Y}{X}$$

Linear D.E.

$$\text{I.F.} = e^{\int \frac{-1}{X} dX} = \frac{1}{X}$$

Now,

$$\frac{1}{X} \frac{dY}{dX} + Y \left(\frac{-1}{X^2} \right) = 0$$

$$\frac{d}{dX} \left(\frac{Y}{X} \right) = 0$$

By intergrating

$$\frac{Y}{X} = C + c$$

$$Y = X^2 + cX$$

$$(y + 4) = (x - 2)^2 + c(x - 2)$$

Assign through (0, 0)

$$\text{Hence, } c = 0$$

$$\text{then, } (y + 4) = (x - 2)^2$$

Which also passes through (5, 5)

38. The area enclosed by the closed curve C given by the differential equation $\frac{dy}{dx} + \frac{x+a}{y-2} = 0$, $y(1) = 0$ is 4π .

Let P and Q be the points of intersection of the curve C and the y-axis. If normals at P and Q on the curve C intersect x-axis at points R and S respectively, then the length of the line segment RS is :

अवकल समीकरण $\frac{dy}{dx} + \frac{x+a}{y-2} = 0$, $y(1) = 0$ से प्राप्त संवृत्त वक्र C से घिरे क्षेत्र का क्षेत्रफल 4π है। माना वक्र C व y अक्ष के प्रतिच्छेदन बिंदु P व Q है। यदि वक्र C के बिंदु P व Q पर अभिलम्ब x अक्ष को क्रमशः बिंदु R व S पर मिलते हैं, तब रेखा खण्ड RS की लम्बाई है :

(A) $\frac{2\sqrt{3}}{3}$

(B) 2

(C) $2\sqrt{3}$

(D) $\frac{4\sqrt{3}}{3}$

Ans. D

Sol. $\frac{dy}{dx} + \frac{x+a}{y-2} = 0$

$$\frac{dy}{dx} = \frac{x+a}{2-y}$$

$$(2-y) dy = (x+a) dx$$

$$2y - \frac{y^2}{2} = \frac{x^2}{2} + ax + c$$

$$a + c = -\frac{1}{2} \text{ as } y(1) = 0$$

$$X^2 + Y^2 + 2ax - 4y - 1 - 2a = 0$$

$$\pi r^2 = 4\pi$$

$$r^2 = 4$$

$$4 = a^2 + 4 + 1 + 2a$$

$$(a+1)^2 = 0$$

$$P \text{ or } Q = (0, 2 \pm \sqrt{3})$$

$$\text{Equation of normal at P, Q are } y - 2 = \sqrt{3}(x - 1)$$

$$y - 2 = -\sqrt{3}(x - 1)$$

$$R = \left(1 - \frac{2}{\sqrt{3}}, 0\right)$$

$$S = \left(1 + \frac{2}{\sqrt{3}}, 0\right)$$

$$RS = \frac{4}{\sqrt{3}} = 4 \frac{\sqrt{3}}{3}$$

39. The value of $\sum_{n=1}^{100} \int_{n-1}^n e^{x-[x]} dx$, where $[x]$ is the greatest integer $\leq x$, is :

$$\sum_{n=1}^{100} \int_{n-1}^n e^{x-[x]} dx, \text{ जहाँ } [x] \text{ महत्तम पूर्णांक } \leq x \text{ है, का मान है :}$$

- (A) $100e$ (B) $100(1 - e)$ (C) $100(1 + e)$ (D) $100(e - 1)$

Ans. D

Sol. $S = \sum_{n=1}^{100} \int_{n-1}^n e^{x-[x]} dx = \sum_{n=1}^{100} \int_{n-1}^n e^{\{x\}} dx.$

Since $\{x\}$ is periodic with

Fundamental period = 1,

$$\begin{aligned} S &= 100 \int_0^1 e^{\{x\}} dx = 100 \int_0^1 e^x dx \\ &= 100 \left[e^x \right]_0^1 \\ &= 100(e-1) \end{aligned}$$

40. If $f(x) = \int \frac{5x^8 + 7x^6}{(x^2 + 1 + 2x^7)^2} dx, (x \geq 0)$ and $f(0) = 0$, then the value of $f(1)$ is :

यदि $f(x) = \int \frac{5x^8 + 7x^6}{(x^2 + 1 + 2x^7)^2} dx, (x \geq 0)$ तथा $f(0) = 0$ है तो $f(1)$ का मान है -

- (A) $\frac{1}{2}$ (B) $-\frac{1}{4}$ (C) $\frac{1}{4}$ (D) $-\frac{1}{2}$

Ans. C

Sol.
$$f(x) = \int \frac{5x^8 + 7x^6}{(x^2 + 1 + 2x^7)^2} dx = \int \frac{5x^8 + 7x^6}{x^{14} \left(\frac{1}{x^5} + \frac{1}{x^7} + 2 \right)^2} dx$$

$$\frac{1}{x^5} + \frac{1}{x^7} + 2 = t$$

$$= \int \frac{-dt}{t^2} = \frac{1}{t} = \frac{1}{2 + \frac{1}{x^5} + \frac{1}{x^7}} + C$$

As $f(0) = 0$ So $f(x) = \frac{x^7}{2x^7 + x^2 + 1}$

$$f(1) = \frac{1}{4}$$